Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer Networking LAB

Course Code: CSE 342.2

Experiment No.: 04

Experiment Name: Implementation and verification of Basic VLAN and Trunking.

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☐ Experiment No: 04

☐ Experiment Name: Implementation and verification of Basic Virtual local area networks (VLANs) and Trunking.

☐ Introduction: A virtual local area network (VLAN) is a logical grouping of network devices that allows administrators to segment a single Physical switch into multiple logical networks. Devices within the same VLAN can communicate as if they were attached to the same Physical wire, regardless of their actual physical location. By default, all Ports on a Cisco switch belong to VLAN 1 (the native and default VLAN).

Implementing VLANs is crucial is modern networking because it reduces the size of broadcast domains, Improves overall network Performance by restricting broadcast traffic to specific groups and enhances security by logically separating sensetive departments. [e.g: separating faculty network for students network]

To allow devices in the same VLAN but located on different Physical switches to communicate a trunk link must be established. A trunk Port uses IEEE 802.1q encapsulation to tag ethernet frames with a VLAN ID allowing traffic from multiple VLANs to traverse a single Physical Connections between switches.

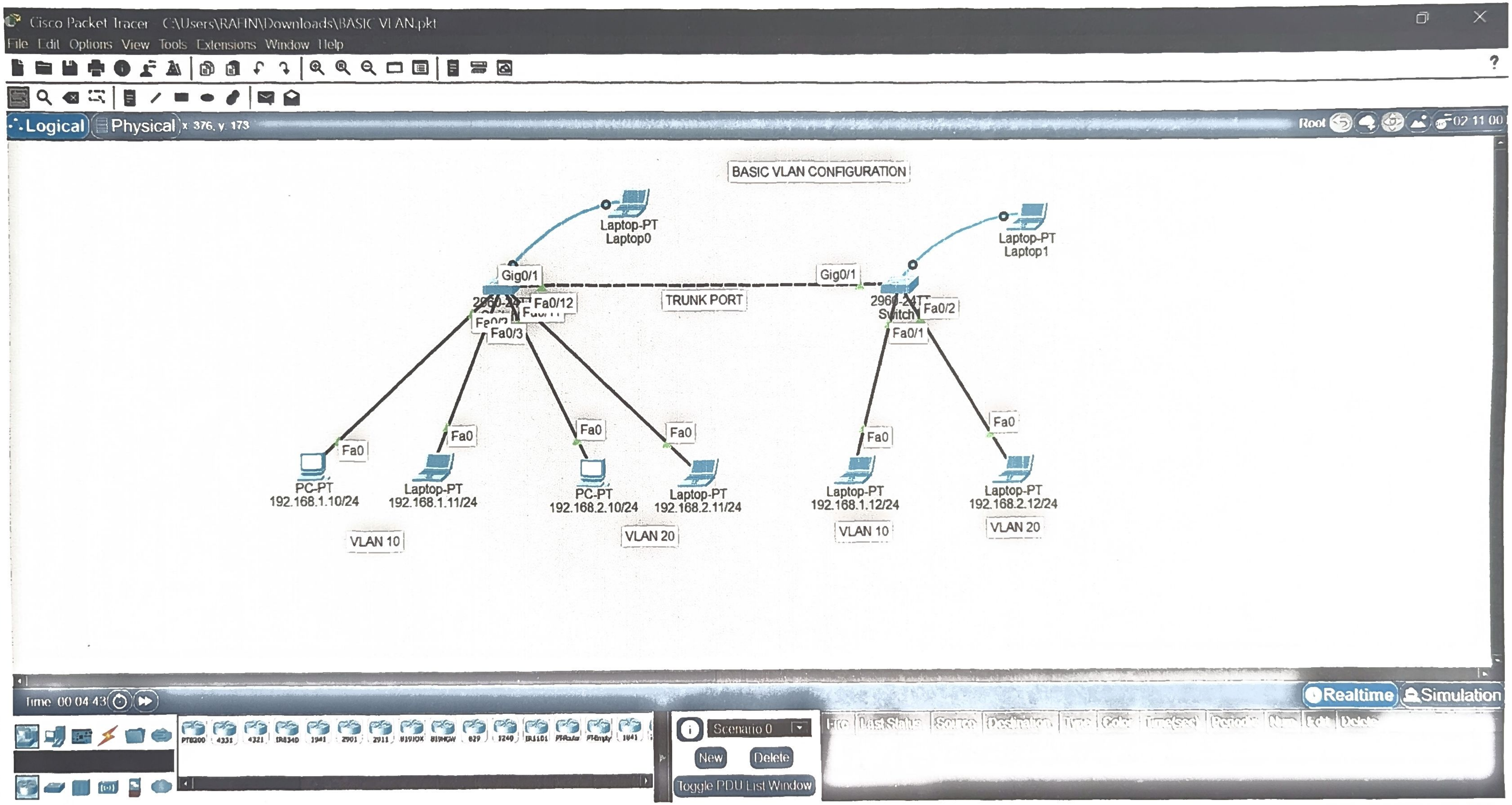
Objective:

- To understand the fundamental concepts of VLAN and their role in network segmentation.
- To design and segment a physical LAN into multiple logical broadcast domains.
- To assign end-device ports to specific VLAN for network isolation.
- To configure an IEEE 802.1Q trunk link between two switches to enable intra-VLAN communication across the network infrastructure.

Network topology and IP addressing:

The network is divided into two separate broadcast domains, ensuring traffic isolation between different device groups.

Device	VLAN ID	End device IP Address	Subnet mask
Switch 0	10	192.168.1.10, 192.168.1.11	255.255.255.0
Switch 0	20	192.168.2.10, 192.168.2.11	255.255.255.0
Switch 1	10	192.168.1.12	255.255.255.0
Switch 1	20	192.168.2.12	255.255.255.0


Cisco Packet Tracer - C:\Users\RAFIN\Downloads\BASIC VLAN.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x 376, y 173
Root
BASIC VLAN CONFIGURATION
Laptop-PT
Laptop0
Laptop-PT
Laptop1
Gig0/1
Gig0/1
TRUNK PORT
Fa0/12
Fa0/3
Fa0/2
Fa0/1
Fa0
PC-PT
192.168.1.10/24
Laptop-PT
192.168.1.11/24
PC-PT
192.168.2.10/24
Laptop-PT
192.168.2.11/24
Laptop-PT
192.168.1.12/24
Laptop-PT
192.168.2.12/24
VLAN 10
VLAN 20
VLAN 10
VLAN 20
Time: 00:04:43
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window

4 **Configuration:** for VLAN creation access the global config mode to establish the logical networks.

```
Switch > enable
Switch# configure terminal
Switch(config)# VLAN 10
Switch(config)# name students
Switch(config)# exit
Switch(config)# VLAN 20
Switch(config)# name faculties
Switch(config)# exit
```

- Bind the fastEthernet interfaces connected to the end devices to their respective VLANs.

```
Switch(config)# interface fa 0/2-3
Switch(config)# switchport mode access
Switch(config)# switchport access VLAN 10
Switch(config)# exit

Switch(config)# interface fa 0/11-12
Switch(config)# switchport mode access
Switch(config)# switchport access VLAN 20
Switch(config)# exit.
```

- Trunk Configuration: Configure the gigabitEthernet link to tag and transport frames from multiple VLANs between the switches. Executed on both switch 0 and switch 1:

```
Switch(config)# Interface gig 0/1
switch(config)# switchport mode trunk
Switch(config)# switchport trunk allowed vlan all
Switch(config)# end.
```

Verification and expected outcomes:

- VLAN Database verification: Execute show vlan brief on both switches. The output must confirm that vlan 10 and 20 are active and that the correct fastEthernet ports are bound to them.

- Trunk status verification: Execute show interfaces trunk. The output must confirm that interface gig 0/1 is in trunking mode and that vlans 10 and 20 are permitted across the link.

- Intra-vlan Connectivity: Initiating an ICMP ping from 192.168.1.10 to 192.168.1.12 will yield a successful reply, validating the trunk link's operation.